EN.553.696:	**Methods in Computational Neuroscience (Fall 2026)**
Instructor:	Mark Kramer (mkramer@jhu.edu)
Course Hours:	Tuesday & Thursday, 9:00-10:15 AM, Maryland 109
Office Hours:	Tuesday 1:30-3 PM, Thursday 12-1:30 PM (Wyman S403)
Textbook:	None
Course Website:	https://mark-kramer.github.io/JHU-553-696
Prerequisites:	Calculus (Differential Equations, Linear Algebra), Computer Science (Programming) or consent of Instructor.

In this course we will introduce mathematical and statistical concepts in neuroscience. We will explore different types of computational models (biological, dynamical, artificial) to mimic observed neural activity. We will develop and applying statistical techniques to characterize observed brain time sereis. Throughout this course, we will focus on brain rhythms.

Course Requirements. Our daily course format will consist of lectures and in-class exercises. Class exercises will typically involve Python programming to implement the concepts discussed. Please attend lectures. If you know in advance that you will need to miss a lecture, please notify me by email.

Exams. There are no exams in this course.

Quizzes. Approximately weekly, you will complete a brief in-class quiz. Quiz topics will cover recent course material. Responses to quiz questions must be yours, without outside assistance. I will drop your lowest quiz score.

Teams. Twice during the semester - at the beginning and approximately halfway - I will randomly assign you to a team of 3 or 4 classmates. I will try to make your team members differ in the first and second halves of the course. You will work with your team to achieve two goals:

- **Challenge Problems.** For each topic, I will assign a list of Challenge Problems that investigate further the material presented. During Topics 1-5 (approximately the first half of the semester), your team will visit my office hours. I will ask you to discuss the solution to one Challenge Problem <u>of my choosing</u>. Your goal is to convince me that you understand the problem, the path to a solution, and an implementation of the solution. I may ask any team member, any question I like related to the Challenge Problem. Your team may visit my office hours at most once per week. Your team may revisit as much as you like throughout the semester.

- **Skill Demonstrations**. Your team may complete skill demonstrations. In a skill demonstration, you will implement a technique we discussed in class and apply it to a new scenario (e.g., a new data set, a new model system). You will deposit the methods to reproduce your skill on the course GitHub repository. You will present your skill to the class (10-minute presentation + 5-minute questions). I will provide a slide template and grading rubric early in the semester. Please let me know at least 1 day before class if your team will present.

Why? Large language models (LLMs) are changing how we think about knowledge and computational skills. Advanced LLMs can provide the code and prose to answer any assessments I design. The purpose of working in teams, answering challenge problems aloud, and presenting your work is to practice valuable human skills. If you use an LLM to guide your work, can you explain what that LLM has done (to a peer, to a collaborator, to yourself)? Neuroscience is interdisciplinary and requires more than LLMs solutions – it requires explaining those answers across disciplinary boundaries and to diverse audiences, working with people you like and dislike. That's (for now) a valuable human skill you will practice here.

Grading

Quizzes	[0 to 10]
N Successful Challenge Problem Visits:	$10N$ with N at most 6 ($N \leq 3$ for Topics 1-5)

N Skill Demonstrations: $15N$ with N at most 3 ($N \leq 2$ for Topics 1-5)

Material to be covered. Below is an outline of the material to be covered. Each topic will consist of lectures and computational exercises. Readings will accompany most topics to provide additional details or more context for the role neuroscience plays in society. Modifications to the reading and schedule may occur throughout the semester.

Topic 0	Sept 1	**Introduction, Programming Habits**
Topic 1	Sept 3, 8, 10	**Perceptron, Backpropagation**
Topic 2	Sept 15, 17	**Hopfield Model***
Topic 3	Sept 22, 24	**Recurrent-Neural Networks** (RNN)
Topic 4	Sept 29, Oct 1	**Integrate & Fire Neuron**
Topic 5	Oct 6, 8	**Hodgkin-Huxley Neuron***
		Team reshuffle
Topic 6	Oct 13, 15, 20	**Analyzing Rhythms** (spectra of fields & spikes)
Topic 7	Oct 27, 29, Nov 3	**Analyzing Coupled Rhythms** (coherence & CFC)
Topic 8	Nov 5, 10	**Gamma Rhythm** (ING, PING, sparse PING)
Topic 9	Nov 12,17	**Beta Rhythm** (funky currents & bursting)
Topic 10	Nov 19	**Not a Rhythm** (aperiodic exponent)
		Fall Recess
Topic ?	Dec 1,3	Buffer / Additional Topics ***Dynamical Models of Spiking Neurons*** ***Sample Size Calculations for Engineers*** ***Regression for Engineers***
	~~Dec 8~~	*(No class due to AES Annual Meeting)*
Topic ?	Dec 10	Buffer / Additional Topics

* Nobel Prize work